Pollution, Transportation Choices, and Health Outcomes

Final Project RDB

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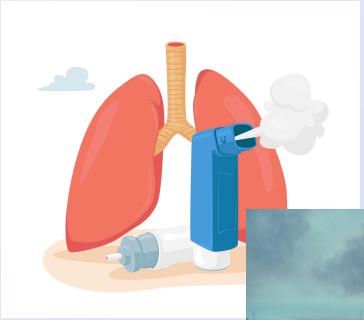
RDB Background & Potential Use Cases

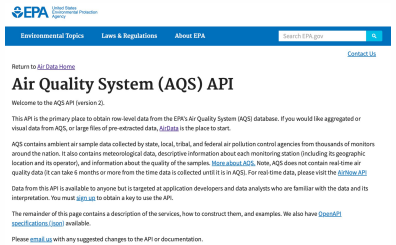
Goal of project was to create an RDB that would allow end users to quickly look at associations between county, cbsa, levels of various pollutants (O2, NO2, PM2.5), actual health outcomes, and transportation patterns in 2022.



E: Data Sources (All Stored on R3)

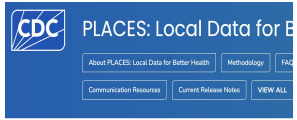
01 EPA Data

- Annual levels of various pollutants collected at various sites
- Includes county/state FIPS codes, cbsa codes & names (note: pm2.5 file no cbsa)
- I queried 3 times separately for NO2, Ozone, and PM2.5 → JSON → R3



02 CDC PLACES Data

- Crude prevalence of asthma, COPD, stroke by county
- Population & 18+ population



03/04 NHTS Data

- Shows *number of trips* made between an origin and destination "zone" in various columns: by distance, mode (air, rail, etc.), purpose (work/nonwork), etc.
- Included a file converting from zone to CBSA

05 Crosswalk Data

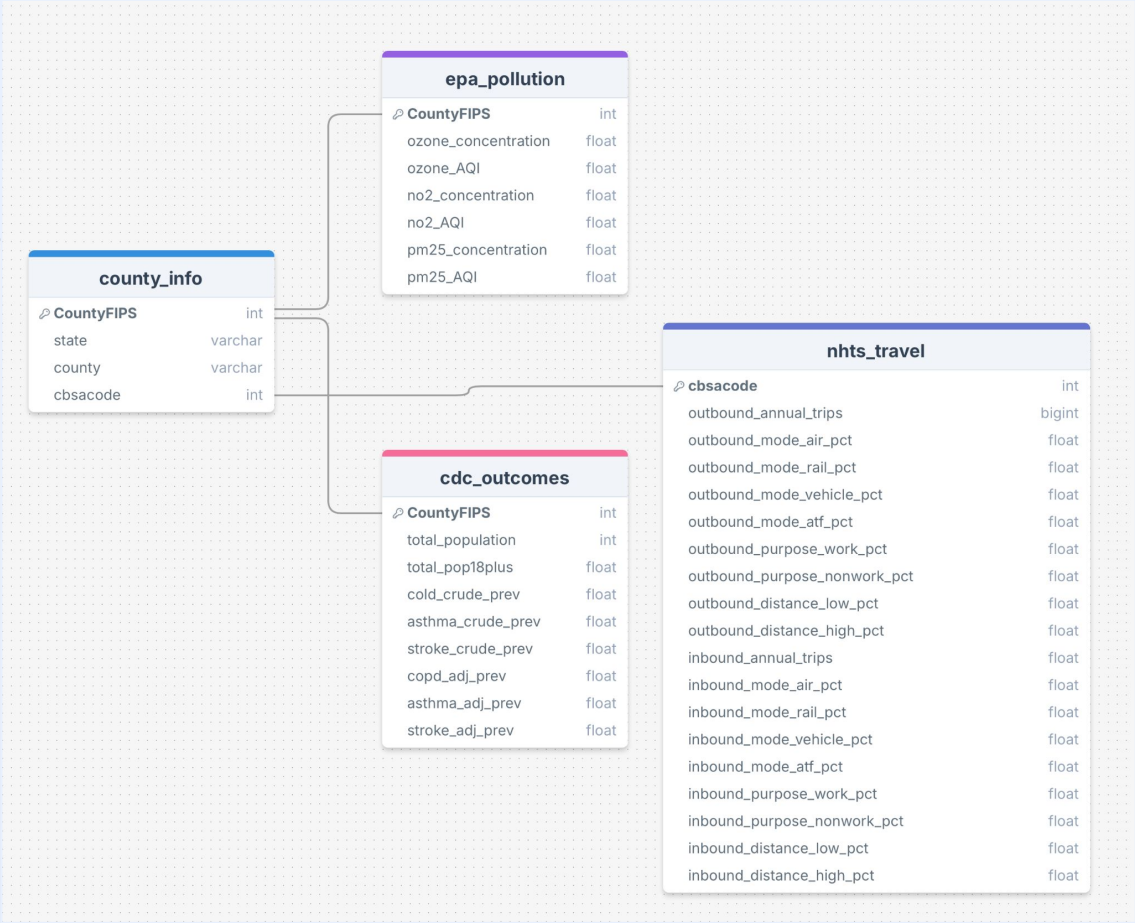
NBER | NATIONAL BUREAU of ECONOMIC RESEARCH

- Converts counties to CBSAs

E: Data Exploration – Cleanup & Filtering

- CDC, Crosswalk, and Lookup mostly clean
- Removed NA rows in EPA data
- Converted to consistent formats (floats to integers)
- Removed duplicate rows (turns out there were duplicates in my EPA data)
- Renamed columns to be consistent across, dropped columns
- Removed flagged rows (would do additional dropping later) in NHTS data
- Flagged rows where CBSA data was missing in EPA data so I could later cross-check with Bureau data
- Liked the CDC format for a singular unified CountyFIPS, so made sure everything else followed the same format

Transformation

- **Overall "why":** I decided to aggregate into countyFIPS (chose to aggregate by county_FIPS so that i could preserve max granularity) and then re-formatted tables so that each row corresponded to a **unique CountyFIPS or cbsa code.**
  - Ex: merged all EPA data, aggregated all site measurements for EACH CountyFIPS into an annual average
  - ^ Aggregated CDC data to get average prevalence, summed populations for each FIPS county & recalculated age-adjusted prevalence
  - NHTS data was aggregated by *cbsacode*, to see inbound/outbound info
- Removed rural areas from NHTS data since they made their own custom, but there was really no way to merge/compare with CDC/EPA - this removed 50% of the data (rural areas were a decent chunk of the data). Also removed rows where there was 0 across the board (likely due to no survey respondents) but this still left us with plenty of data (150,382+ rows).
- Merged CBSA codes, prioritizing that of the Bureau and then falling back on ones from EPA in case data was missing.

L: Updated Schema